Week 5:
Please follow the steps shown by Sir to create and use your Azure account.

Demo Volumes:
```
%sql
create volume deltalake_catalog.default.managed_vol;
```

Python Code:
spark.sql("CREATE VOLUME deltalake_catalog.default.managed_vol")

%fs
mkdirs /Volumes/deltalake_catalog/default/managed_vol/emp

```
%sql
select * from
CSV.`/Volumes/deltalake_catalog/default/managed_vol/emp`
```

Python code:
```
df = (spark.read
      .format("csv")
      .option("header", "true")
      .load("/Volumes/deltalake_catalog/default/managed_vol/emp"))

display(df)
```

```
df = spark.read.format("csv").option("header",
True).load("/Volumes/deltalake_catalog/default/managed_vol/emp")

display(df
```

```
%sql
drop volume deltalake_catalog.default.managed_vol;
```

Python code:
spark.sql("DROP VOLUME deltalake_catalog.default.managed_vol")

```
%sql
create volume deltalake_catalog.default.managed_vol;
```

Python code:
spark.sql("CREATE VOLUME deltalake_catalog.default.managed_vol")

```
df = spark.read.format("csv").option("header",
True).load("/Volumes/deltalake_catalog/default/managed_vol/emp")
```

```
%sql
create external volume deltalake_catalog.default.external_vol
LOCATION '/Workspace/Users/technicalsupport@trendytech.in/ext_volume'
```

Python Code:

```
spark.sql("""
CREATE EXTERNAL VOLUME deltalake_catalog.default.external_vol
LOCATION '/Workspace/Users/technicalsupport@trendytech.in/ext_volume'
""")
```

```
%sql
drop volume deltalake_catalog.default.external_vol;
```

Python code:

```
spark.sql("DROP VOLUME deltalake_catalog.default.external_vol")
```

```
%sql
create external volume deltalake_catalog.default.external_vol
LOCATION
'abfss://externaldata@ttmystorageaccount001.dfs.core.windows.net/ext_volume'
```

Python code:

```
spark.sql("""
CREATE EXTERNAL VOLUME deltalake_catalog.default.external_vol
LOCATION 'abfss://externaldata@ttmystorageaccount001.dfs.core.windows.net/ext_volume'
""")
```

```
%sql
select * from CSV.`/Volumes/deltalake_catalog/default/external_vol/emp`
```

Python code:

```
df = spark.read.format("csv").option("header",
True).load("/Volumes/deltalake_catalog/default/external_vol/emp")

display(df)
```

CTAS:

```
%sql
use catalog deltalake_catalog;
```

Python code:
spark.sql("USE CATALOG deltalake_catalog")

```
%sql
create volume deltalake_catalog.default.raw;
```

Python code:
spark.sql("CREATE VOLUME deltalake_catalog.default.raw")

%fs
mkdirs /Volumes/deltalake_catalog/default/raw/ordershistory

```
%sql
LIST '/Volumes/deltalake_catalog/default/raw/ordershistory'
```

Python Code:
dbutils.fs.ls("/Volumes/deltalake_catalog/default/raw/ordershistory")

%fs
ls '/Volumes/deltalake_catalog/default/raw/ordershistory'

```
%sql
select * from CSV.`/Volumes/deltalake_catalog/default/raw/ordershistory`
```

Python Code:
df = spark.read.format("csv").option("header", "true").load("/Volumes/deltalake_catalog/default/raw/ordershistory")
display(df)

```
%sql
drop table if exists orders;

create table orders
as
select * from
read_files(
  '/Volumes/deltalake_catalog/default/raw/ordershistory',
  format => 'csv'
)
limit 10;
```

Python Code:
spark.sql("DROP TABLE IF EXISTS orders")

```
df = (spark.read
      .format("csv")
      .option("header", "true")
      .option("inferSchema", "true")
      .load("/Volumes/deltalake_catalog/default/raw/ordershistory")
      .limit(10))

df.write.saveAsTable("orders")
```

```
%sql
select * from orders;
```

Python code:
spark.sql("SELECT * FROM orders").show()

```
%sql
describe formatted orders;
```

Python code:
spark.sql("DESCRIBE FORMATTED orders").show(truncate=False)

```
%sql
drop table if exists orders;

create table orders
as
select * from
read_files(
  '/Volumes/deltalake_catalog/default/raw/ordershistory',
  format => 'csv',
  schema => 'order_id BIGINT, order_date STRING, customer_id BIGINT,
order_status STRING, _rescued_data STRING',
  header => true,
  mode => 'PERMISSIVE',
  columnNameofCorruptRecord => '_rescued_data'
```

```
)
```

Python code:
spark.sql("DROP TABLE IF EXISTS orders")

from pyspark.sql.types import StructType, StructField, LongType, StringType

schema = StructType([
    StructField("order_id", LongType(), True),
    StructField("order_date", StringType(), True),
    StructField("customer_id", LongType(), True),
    StructField("order_status", StringType(), True),
    StructField("_rescued_data", StringType(), True)
])

df = (spark.read
      .format("csv")
      .schema(schema)
      .option("header", "true")
      .option("mode", "PERMISSIVE")
      .option("columnNameOfCorruptRecord", "_rescued_data")
      .load("/Volumes/deltalake_catalog/default/raw/ordershistory"))

df.write.saveAsTable("orders")

Copy into demo-1

```
%sql
use catalog deltalake_catalog;
```

Python Code:
spark.sql("USE CATALOG deltalake_catalog")

```
%sql
drop table if exists orders;
create table orders;
```

Python code:

spark.sql("DROP TABLE IF EXISTS orders")

spark.sql("CREATE TABLE orders ()")

```
%sql
copy into deltalake_catalog.default.orders
from "/Volumes/deltalake_catalog/default/raw/ordershistory"
FILEFORMAT = CSV
FORMAT_OPTIONS(
  'inferSchema' = 'true',
  'header' = 'true',
  'mergeSchema' = 'true',
  'delimiter' = ','
)
COPY_OPTIONS (
  'mergeSchema' = 'true'
);
```

Python Code:

```
spark.sql("""
COPY INTO deltalake_catalog.default.orders
FROM '/Volumes/deltalake_catalog/default/raw/ordershistory'
FILEFORMAT = CSV
FORMAT_OPTIONS (
 'inferSchema' = 'true',
 'header' = 'true',
 'mergeSchema' = 'true',
 'delimiter' = ','
)
COPY_OPTIONS (
 'mergeSchema' = 'true'
)
""")
```

```
%sql
select * from deltalake_catalog.default.orders;
```

Python Code:

```
df = spark.sql("SELECT * FROM deltalake_catalog.default.orders")
df.show()
```

```
%sql
describe formatted deltalake_catalog.default.orders;
```

Python Code:

```
df = spark.sql("DESCRIBE FORMATTED deltalake_catalog.default.orders")
df.show(truncate=False)
```

```
%sql
describe history deltalake_catalog.default.orders;
```

Python Code:
df = spark.sql("DESCRIBE HISTORY deltalake_catalog.default.orders")
df.show(truncate=False)

Copy-into demo-2:

```
%sql
use catalog deltalake_catalog;
```

Python code:
spark.sql("USE CATALOG deltalake_catalog")

```
%sql
drop table if exists orders;
```

Python code:
spark.sql("DROP TABLE IF EXISTS orders")

```
%sql
create table orders(
  order_id int,
  order_date string,
  customer_id int,
  order_status string,
  _file_name string,
  _load_date timestamp
) using delta;
```

Python code:
```
spark.sql("""
CREATE TABLE orders (
  order_id INT,
  order_date STRING,
  customer_id INT,
  order_status STRING,
  _file_name STRING,
  _load_date TIMESTAMP
```

) USING DELTA
""")

```
%sql
copy into deltalake_catalog.default.orders
from (select cast(order_id as int) as order_id,
order_date,
cast(customer_id as int) as customer_id,
order_status,
_metadata.file_name as _file_name,
current_timestamp() as _load_date

from "/Volumes/deltalake_catalog/default/raw/ordershistory")
FILEFORMAT = CSV
FORMAT_OPTIONS(
  'header' = 'true'
  ) ;
```

Python code:
spark.sql("""
COPY INTO deltalake_catalog.default.orders
FROM (
  SELECT
    CAST(order_id AS INT) AS order_id,
    order_date,
    CAST(customer_id AS INT) AS customer_id,
    order_status,
    _metadata.file_name AS _file_name,
    current_timestamp() AS _load_date
  FROM '/Volumes/deltalake_catalog/default/raw/ordershistory'
)
FILEFORMAT = CSV
FORMAT_OPTIONS (
  'header' = 'true'
)
""")

```
%sql
select * from deltalake_catalog.default.orders;
```

Python code:
df = spark.sql("SELECT * FROM deltalake_catalog.default.orders")
display(df)

```
%sql
describe formatted deltalake_catalog.default.orders;
```

Python code:
df = spark.sql("DESCRIBE FORMATTED deltalake_catalog.default.orders")
display(df)

```
%sql
describe history deltalake_catalog.default.orders;
```

Python code:
df = spark.sql("DESCRIBE HISTORY deltalake_catalog.default.orders")
display(df)

Autoloader Demo - 1

```
%sql
use catalog deltalake_catalog;
```

Python Code:
spark.sql("USE CATALOG deltalake_catalog")

```
landing_zone = "/Volumes/deltalake_catalog/default/raw"
orders_data = landing_zone + "/ordershistory"
checkpoint_path = landing_zone + "/orders_checkpoint"
```

```
ordersdf = (spark.readStream
.format("cloudFiles")
.option("cloudFiles.format", "csv")
.option("cloudFiles.inferSchema", "true")
.option("cloudFiles.inferColumnTypes", "true")
.option("cloudFiles.schemaLocation", checkpoint_path)
.load(orders_data))
```

Above code is already in pyspark

```
%sql
drop table if exists deltalake_catalog.default.ordersdelta;
```

Python Code:

spark.sql("DROP TABLE IF EXISTS deltalake_catalog.default.ordersdelta")

```
ordersdf.writeStream \
.format("delta") \
.option("checkpointLocation", checkpoint_path) \
.option("mergeSchema", True) \
.outputMode("append") \
.trigger(availableNow=True) \
.toTable("deltalake_catalog.default.ordersdelta")
```

```
%sql
select * from deltalake_catalog.default.ordersdelta;
```

Python Code:
orders_df = spark.table("deltalake_catalog.default.ordersdelta")
orders_df.show()

Autoloader Demo-2,3,4,5

Code is in the pyspark